

RAJALAKSHMI ENGINEERING COLLEGE
THANDALAM, CHENNAI-602105

CYBER AND DIGITAL FORENSICS
LAB MANUAL

Submitted by
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241901027

in partial fulfillment of the award of the degree of

BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING
(CYBER SECURITY)

Exp No:1 M.Dhivya

16/07/2026 241901027

STUDY OF COMPUTER FORENSICS AND DIFFERENT TOOLS USED FOR FORENSIC INVESTIGATION

AIM:

To study the concept of computer forensics, understand the different phases of forensic investigation, and identify various forensic tools and their uses.

PROCEDURE:

1. Study the definition, goals, and importance of computer forensics.
2. Understand the different phases of the forensic process: identification, preservation, collection, examination, analysis, and reporting.
3. Identify commonly used forensic tools such as FTK Imager, Autopsy, USBDeview, ChromeHistoryView, ExifReader, DumpIt, Compare It, and Command Prompt.
4. Study the purpose, features, supported platforms, and forensic applications of each tool.
5. Explore the interfaces of selected tools and observe their basic operations.
6. Record the observations and compare the tools based on their features and uses.
7. Prepare a report summarizing the tools studied and their importance in forensic investigation.

TOOLS STUDIED:

S. No.	Tool	Purpose / Observation
1	FTK Imager	Creates forensic images and supports evidence acquisition and preview.
2	Autopsy	Provides a graphical interface for examining disk images and analyzing digital evidence.
3	USBDeview	Displays information about USB devices connected to a Windows computer.
4	ChromeHistoryView	Displays available browsing-history records from Google Chrome.
5	ExifReader	Reads metadata embedded in supported image files, such as camera details and timestamps when available.
6	DumpIt	A memory-acquisition utility used to capture a system memory dump; use only with proper authorization.
7	Compare It	Compares text or files and helps identify differences between versions.
8	Command Prompt	Provides command-line utilities for basic system inspection and file-related operations.

OBSERVATIONS:

S. No.	Observation area	Observation
1	Evidence acquisition	FTK Imager can be used to acquire or preview forensic evidence.
2	Disk examination	Autopsy organizes and examines data from supported evidence sources.
3	Device history	USBDeview can display available records about USB devices.
4	Browser activity	ChromeHistoryView can display available Chrome history information.
5	Metadata	ExifReader can reveal metadata stored in supported image files.
6	Memory evidence	DumpIt can capture memory contents when run in an appropriate, authorized environment.
7	File comparison	Compare It helps identify differences between two files.
8	Command-line operations	Command Prompt provides basic commands for system and file inspection.

RESULTS:

The concept and phases of computer forensics were studied, and different forensic investigation tools and their purposes were identified. Their importance in collecting, examining, and analyzing digital evidence was understood

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LIVE FORENSIC CASE INVESTIGATION USING AUTOPSY

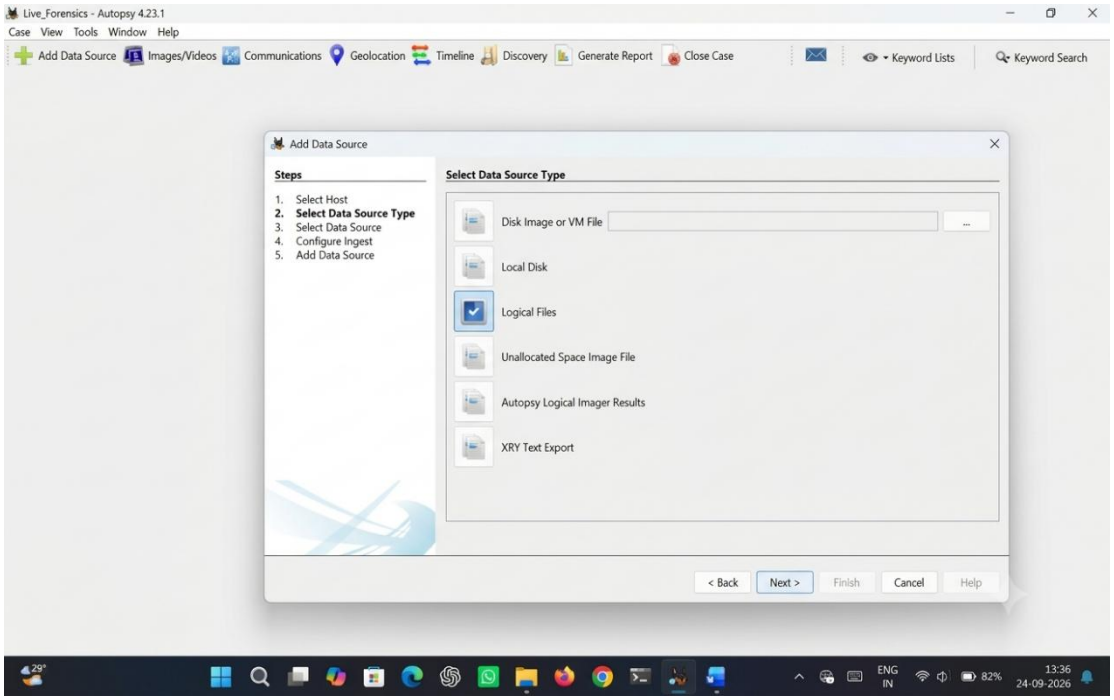
AIM:

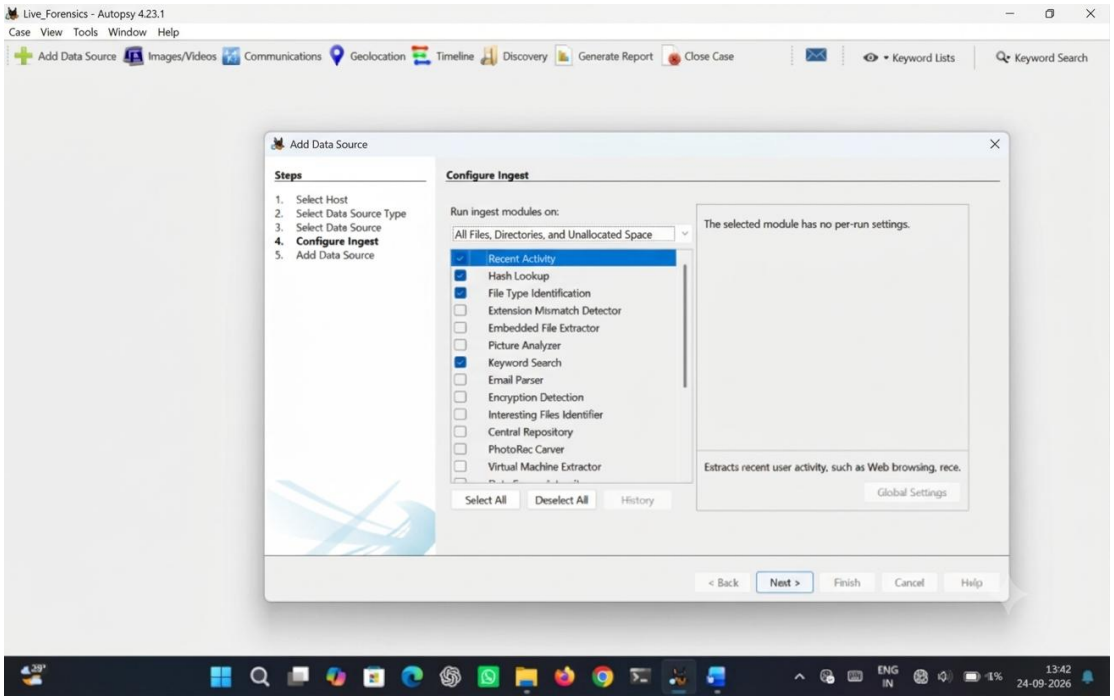
To perform a live forensic investigation using Autopsy and identify, extract, and analyze digital evidence, including system information, user accounts, recent documents, web history, cookies, downloads, and metadata.

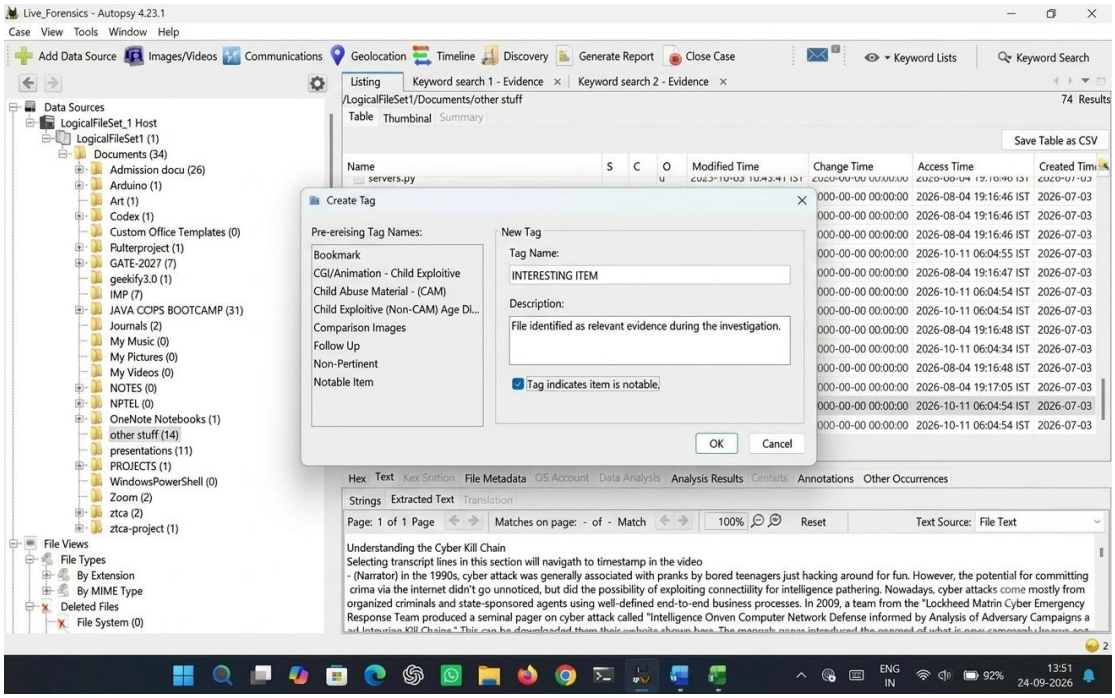
PROCEDURE:

1. Download and install the Autopsy forensic tool on the computer.
2. Open Autopsy and select New Case. Enter the case name, base directory, case number, and examiner details.
3. Add the data source by selecting the source type and browsing to the required disk image or evidence source.
4. Configure the ingest modules and select the required modules for a complete forensic analysis.
5. Click Finish to begin processing the data source and storing the results in the local database.
6. After processing is complete, examine the forensic investigation results.
7. Select Devices Attached to view the devices connected to the system.
8. Examine EXIF Metadata, Installed Programs, and Operating System Information to collect system-related evidence.
9. Select Operating System User Accounts to identify the user accounts available on the system.
10. Open Recent Documents to view recently created or accessed documents.
11. Examine Web Bookmarks, Web Cookies, Web Downloads, Web History, and Web Search to analyze browser-related evidence.
12. Select Email Addresses to identify email addresses found in the evidence.
13. Review the extracted information and document the relevant forensic findings.

OBSERVATIONS:







RESULTS:

Thus, the live forensic case investigation was performed using Autopsy, and the available digital evidence was extracted and analyzed successfully.

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FTK IMAGER

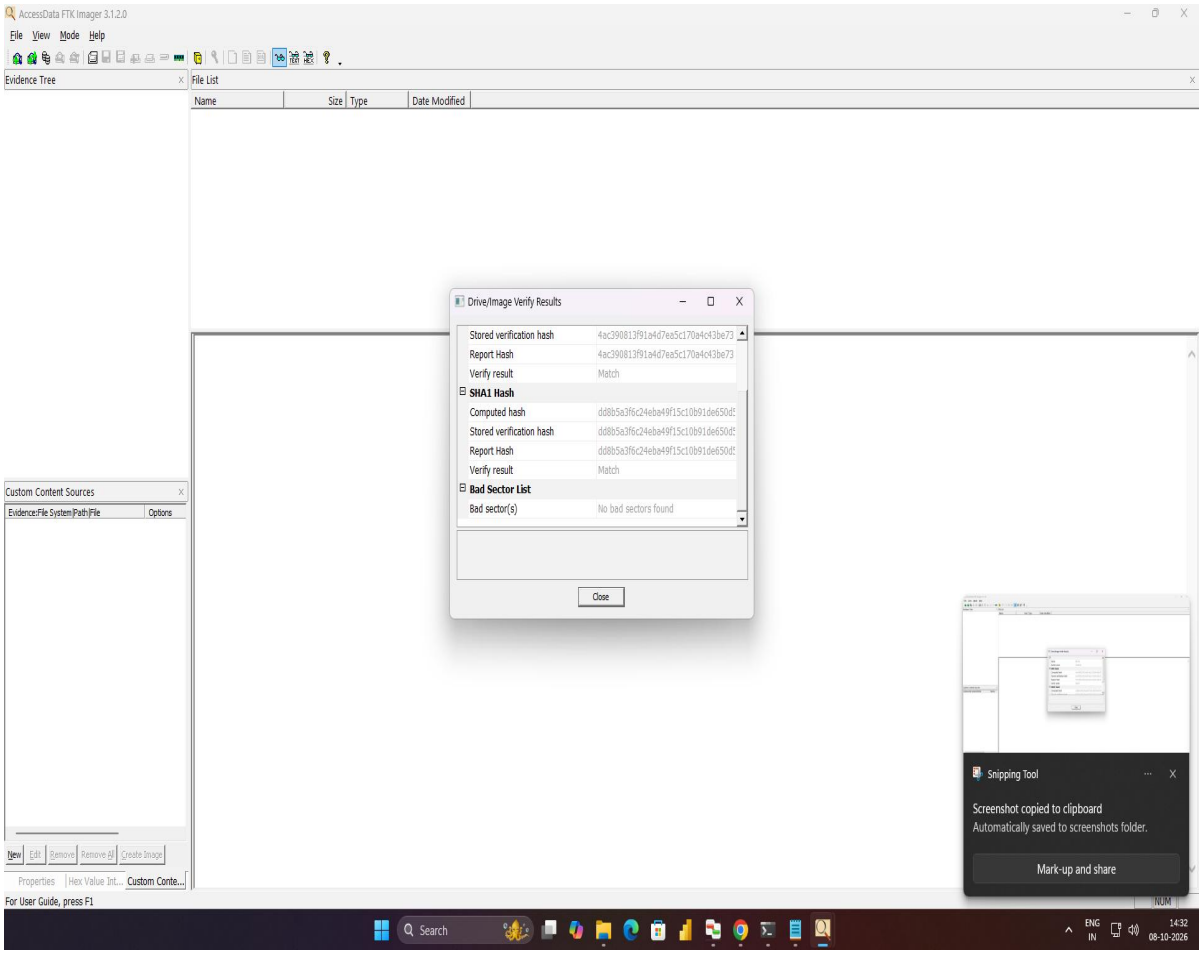
AIM:

To acquire a forensic image of a storage device using FTK Imager while preserving the integrity of digital evidence and verifying it using hash values.

PROCEDURE:

1. Open FTK Imager and select Create Disk Image.
2. Select the source evidence type, such as Physical Drive, Logical Drive, Image File, or Folder Contents.
3. Choose the required source drive and click Finish.
4. Select the destination image format as E01 (Expert Witness Format).
5. Specify the destination folder and enter the case details, including case number, evidence number, examiner name, and description.
6. Select the options to verify the image after acquisition and calculate hash values such as MD5 and SHA-1.
7. Click Start to begin the forensic imaging process.
8. Wait until the image acquisition is completed successfully.
9. Verify the generated hash values and confirm that the image integrity has been maintained.
10. Save the generated forensic image and acquisition report for further investigation.

OBSERVATIONS:



RESULTS:

A forensic image of the storage device was successfully acquired using FTK Imager in E01 format. The hash values were generated and verified to ensure the integrity and authenticity of the digital evidence.

Exp No:4 M.Dhivya

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FIND LAST CONNECTED USB ON YOUR SYSTEM

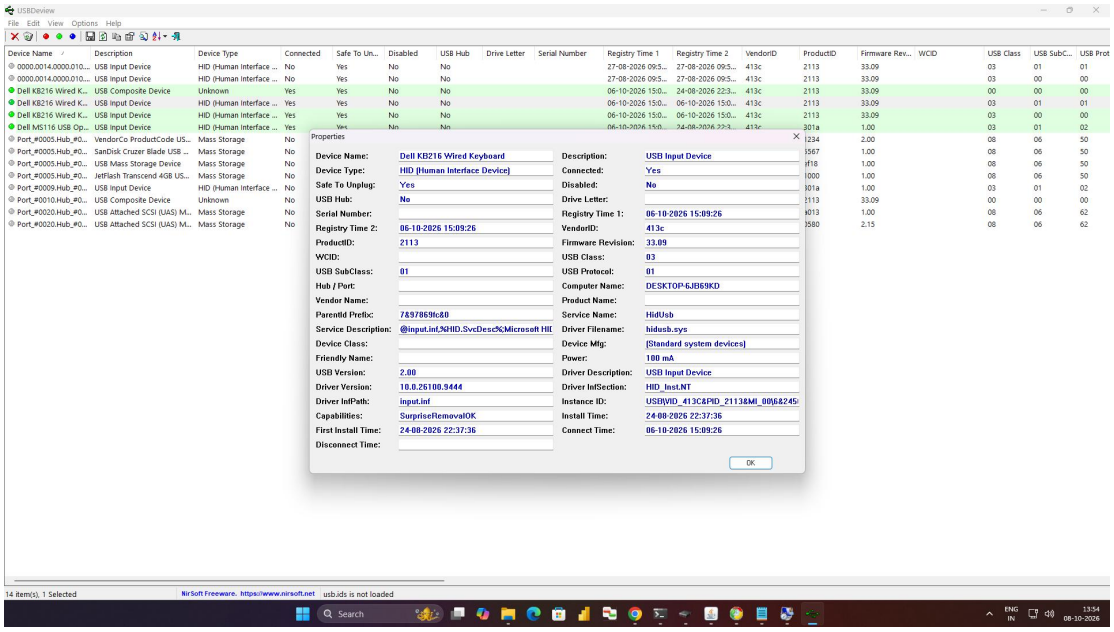
AIM:

To identify the last connected USB device on the system using USBDeview and examine its details, including device name, description, serial number, and last plug/unplug date and time.

PROCEDURE:

1. Open a web browser and visit the NirSoft website.
2. Download the USBDeview ZIP file and extract its contents.
3. Right-click `USBDeview.exe` and select Run as administrator.
4. The USBDeview main window displays a list of currently connected and previously used USB devices.
5. Locate the Last Plug/Unplug Date column to identify the most recently connected or disconnected USB device.
6. Select the required USB device and note its device name, description, serial number, and last plug/unplug date and time.
7. Double-click the USB device to view its detailed information.
8. If required, export the device information using the File → Save Selected Items option.
9. Record the identified USB device details in the lab manual for documentation.

OBSERVATIONS:



RESULTS:

The last connected USB device was identified successfully using USBDeview. Its device name, description, serial number, and last plug/unplug date and time were examined and recorded for USB forensic investigation.

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HOW TO VIEW LAST ACTIVITY OF YOUR PC

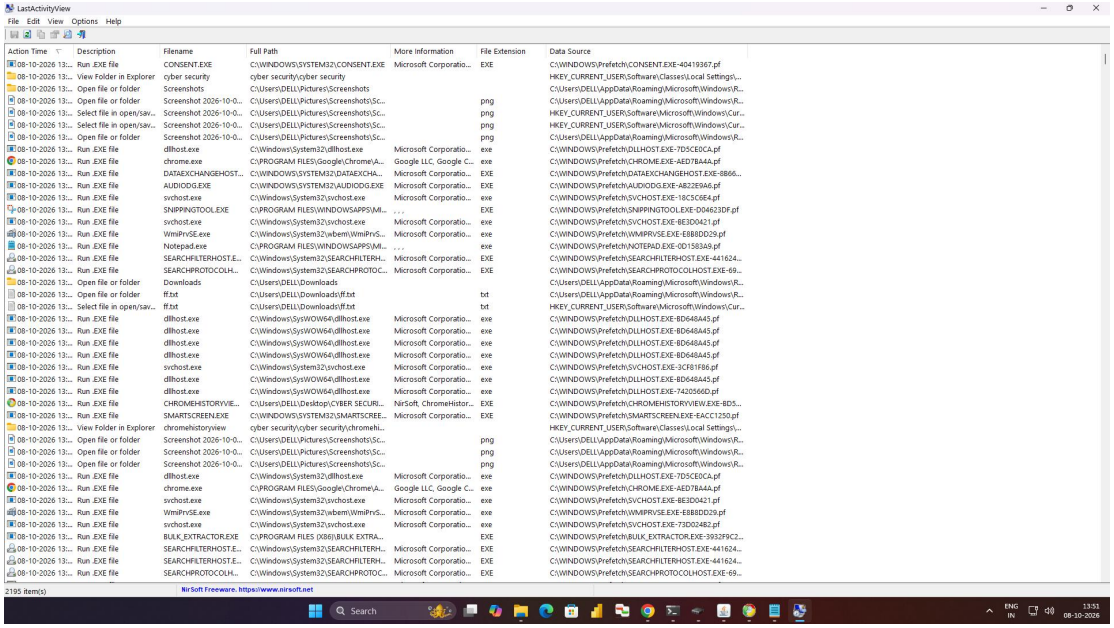
AIM:

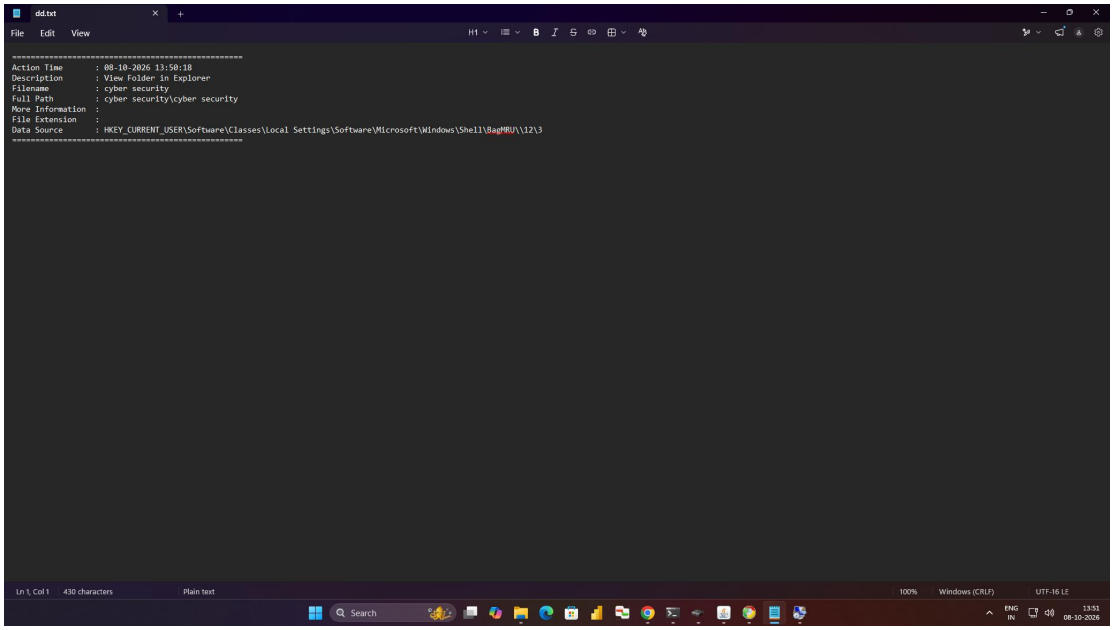
To view and analyze the last activity of a Windows PC using LastActivityView (NirSoft) and examine the recorded activities, including their date, time, description, and file paths.

PROCEDURE:

1. Open a web browser and visit the NirSoft website.
2. Download the LastActivityView ZIP file.
3. Extract the downloaded ZIP file using WinRAR, 7-Zip, or Windows File Explorer.
4. Open the extracted folder and run `LastActivityView.exe`. No installation is required.
5. The main window displays the recent activities recorded on the system.
6. Examine the activity details, including Action Time, Entry Type, Description, and Filename/Full Path.
7. Select an activity and double-click it to view detailed information.
8. If required, use the filter option to view specific types of activities.
9. Save the activity report using the File → Save Selected Items option in a supported format, such as TXT, CSV, or HTML.
10. Record the important observations, analyze the collected information, and close the application after completing the investigation.

OBSERVATIONS:





RESULTS:

The recent activities of the Windows PC were viewed and analyzed successfully using LastActivityView. The recorded activity details, including date, time, description, and file paths, were examined and documented for digital forensic investigation.

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HOW TO EXTRACTING BROWSER ARTIFACTS

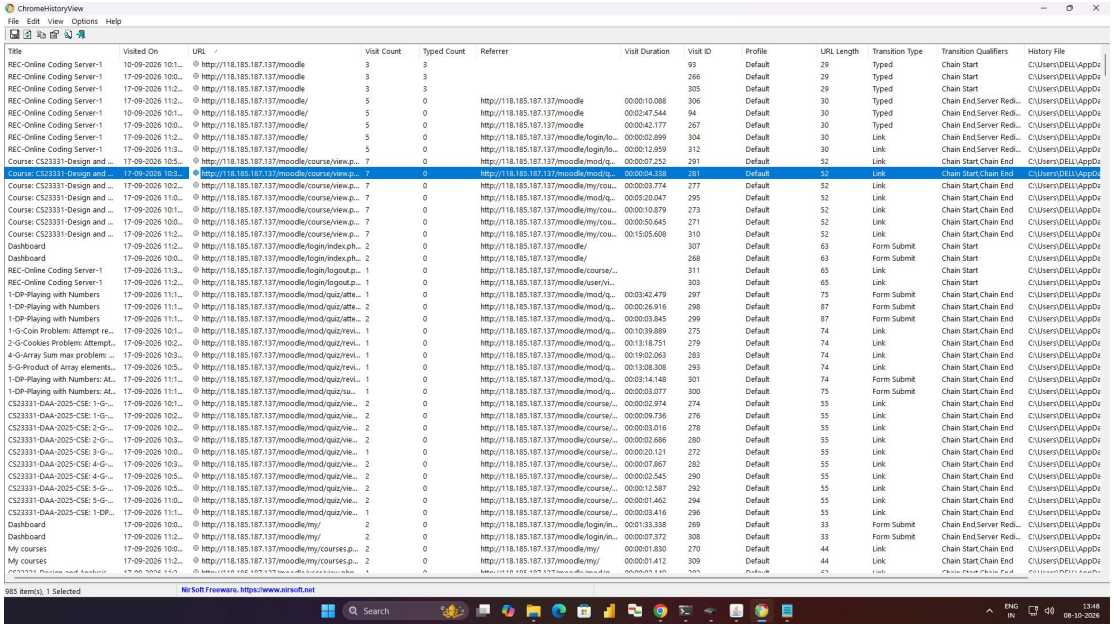
AIM:

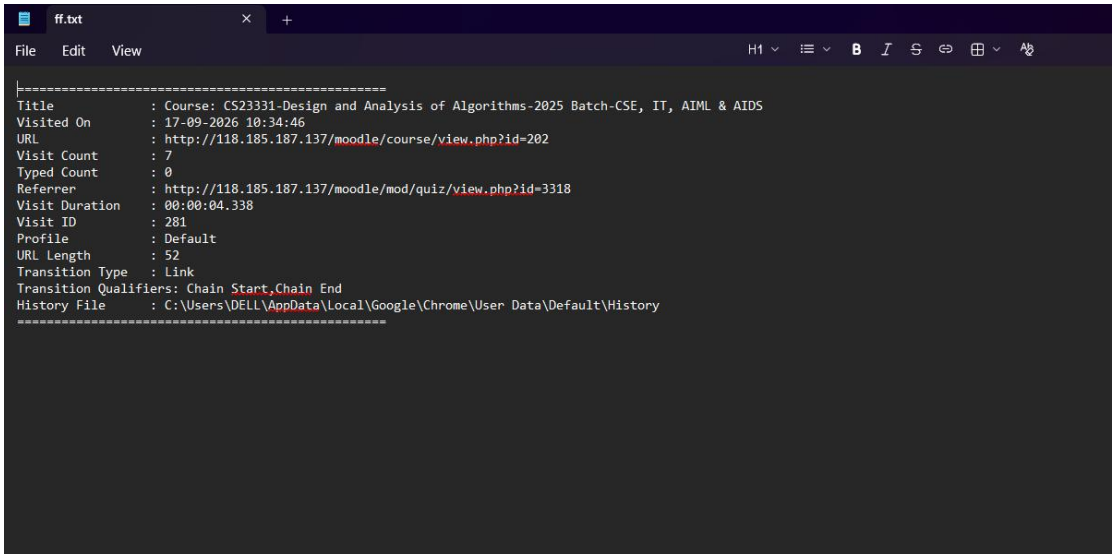
To extract and analyze browser artifacts, such as visited URLs, page titles, visit dates and times, and other browsing history details using ChromeHistoryView (NirSoft).

PROCEDURE:

1. Open a web browser and visit the NirSoft website.
2. Search for ChromeHistoryView and download the ZIP file.
3. Extract the downloaded ZIP file using WinRAR or 7-Zip into a folder.
4. Open the extracted folder and double-click `ChromeHistoryView.exe`. No installation is required.
5. The tool automatically loads the available Google Chrome browsing history.
6. Examine the list of visited web pages displayed in the main window.
7. Observe the details, including URL, page title, visit date and time, number of visits, typed count, referrer, and visit ID.
8. Open the View menu to select additional columns if required.
9. Review the browsing history to identify frequently visited websites, search activity, and potentially relevant URLs.
10. Optionally, use the File menu to export the collected information in CSV, HTML, or text format for further analysis.
11. Record the observations and important findings in the laboratory record.
12. Close ChromeHistoryView after completing the analysis.

OBSERVATIONS:





RESULTS:

Thus, browser artifacts were extracted and analyzed using ChromeHistoryView, and details such as visited URLs, page titles, visit dates and times, and browsing patterns were examined for digital forensic investigation.

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COMPARISON OF TWO FILES FOR FORENSIC INVESTIGATION USING COMPARE IT SOFTWARE

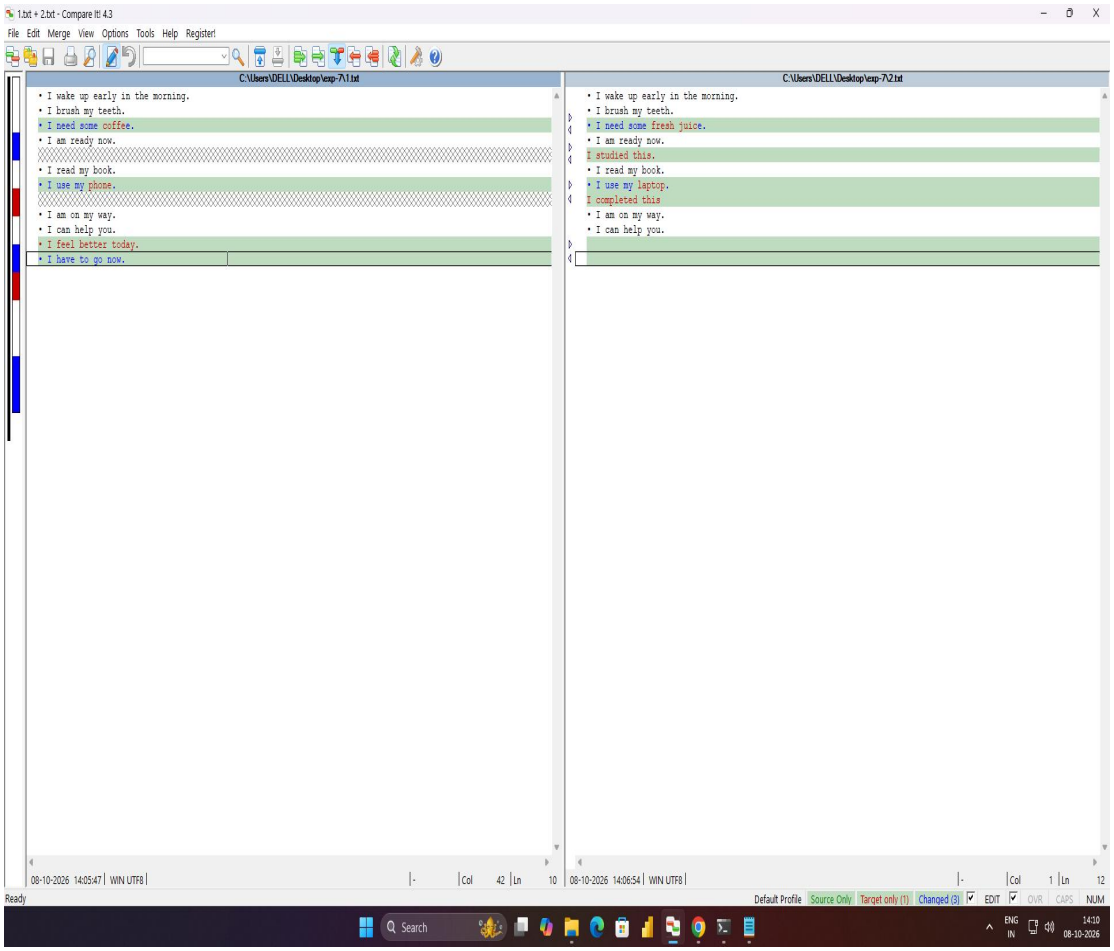
AIM:

To compare two files using Compare It software and identify similarities, differences, additions, deletions, and modifications in their contents for forensic investigation.

PROCEDURE:

1. Download and install the Compare It software on the computer.
2. Open the Compare It application.
3. Select the first file to be examined as the original file.
4. Select the second file to be examined as the modified or suspected file.
5. Load both files into the software for comparison.
6. Start the comparison process using the software's compare option.
7. Observe the contents of both files displayed side by side.
8. Identify the highlighted differences, including added text, deleted text, and modified content.
9. Examine the differences carefully to determine which portions of the files have changed.
10. Record the observations and note the differences identified between the two files.
11. Save or export the comparison results, if the software version supports this option.
12. Close the application after completing the analysis.

OBSERVATIONS:



RESULTS:

Thus, two files were compared successfully using Compare It software, and the similarities and differences between their contents were identified and documented for forensic investigation.

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HOW TO COLLECT EMAIL EVIDENCE IN VICTIM PC

AIM:

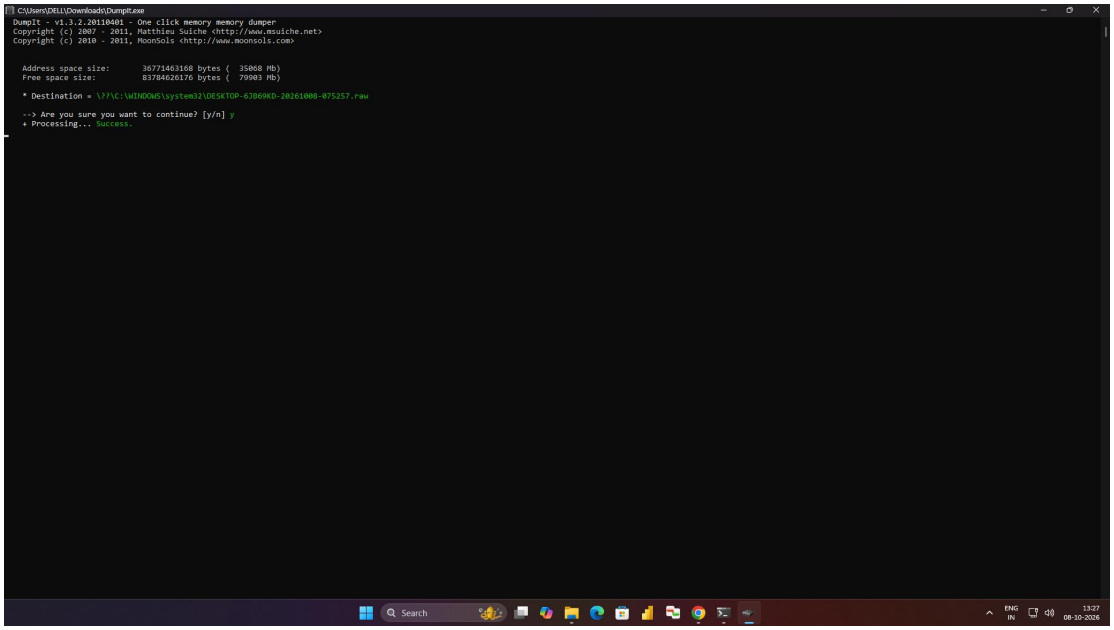
To collect and preserve email-related digital evidence from a victim's computer for forensic investigation without altering the original evidence.

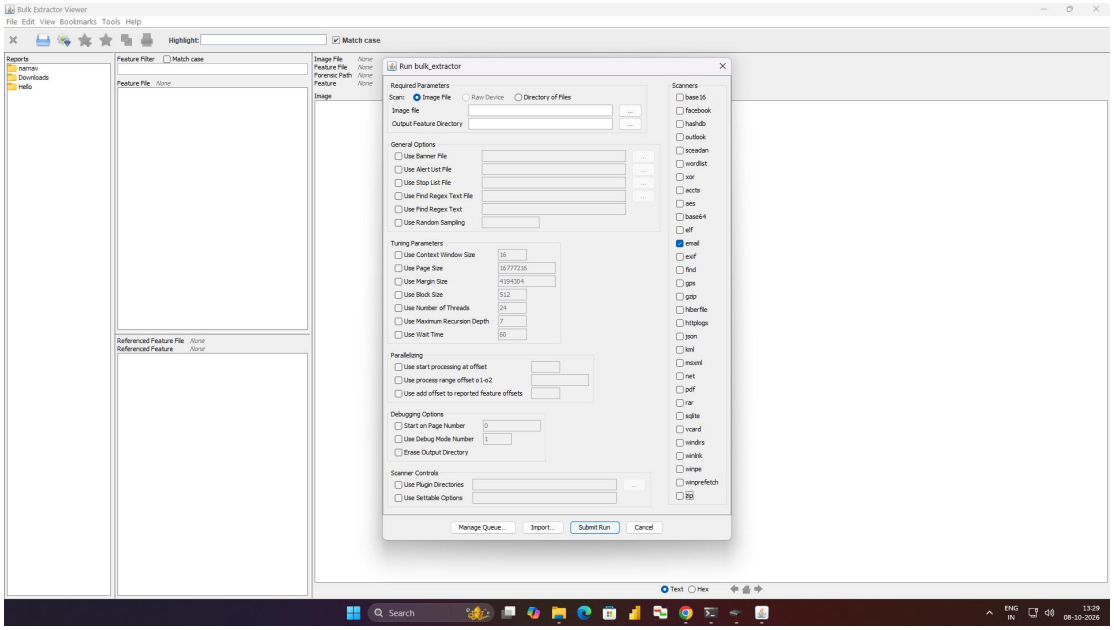
PROCEDURE:

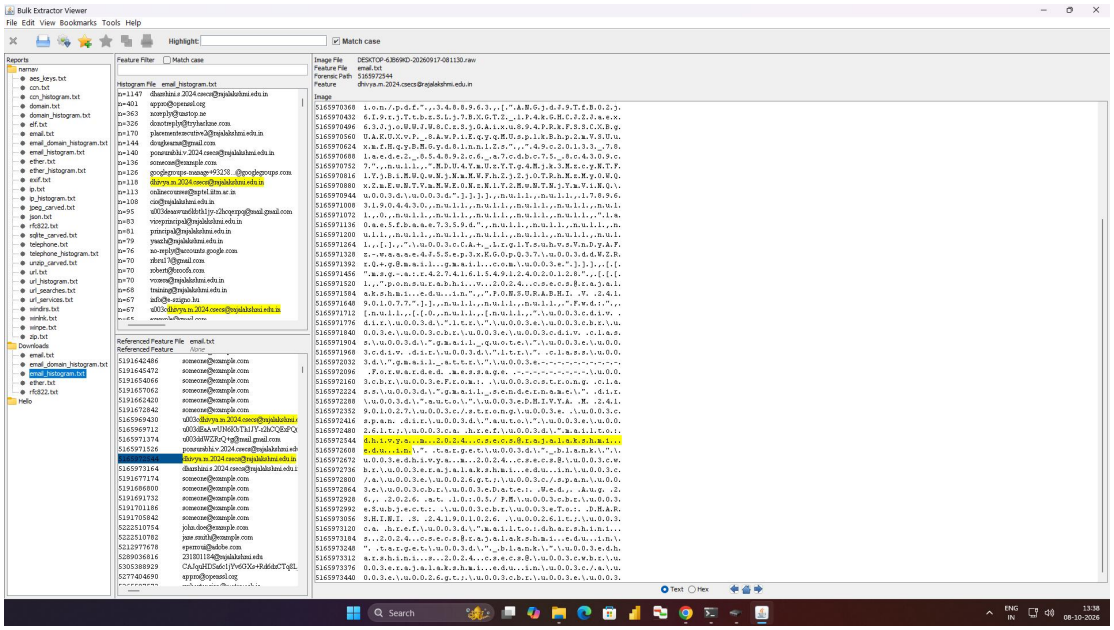
1. Switch on the computer and document its condition, date, time, and other relevant details.
2. Obtain proper authorization before examining the computer.
3. Identify the email application used on the computer, such as Microsoft Outlook or a web browser.
4. Examine the available email accounts and identify relevant emails, folders, attachments, and message headers.
5. If emails are accessed through a web browser, examine the available browser history for email-related websites and activity.
6. If Microsoft Outlook is used, identify relevant email data files, such as `.pst` and `.ost`, where available.
7. Create a forensic copy of the relevant email data using an appropriate forensic acquisition tool.
8. Calculate cryptographic hash values, such as SHA-256, for the acquired evidence to help verify its integrity.
9. Examine email details, including the sender, recipient, subject, date, time, message headers, and attachments.
10. Export relevant emails in suitable forensic formats, such as `.eml` or `.msg`, where supported.
11. Record the evidence details, file locations, hash values, and collection date in an evidence log.

12. Store the collected evidence securely and perform further analysis on working copies rather than the original data.

OBSERVATIONS:







RESULTS:

Thus, email-related digital evidence was collected and documented from the victim's computer using forensic procedures. The relevant email messages, headers, attachments, and associated artifacts were identified and preserved for further investigation.

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HIDE AND EXTRACT ANY TEXT FILE BEHIND AN IMAGE FILE USING COMMAND PROMPT

AIM:

To hide a text file behind an image file and extract the hidden text file using Windows Command Prompt.

PROCEDURE:

Part A: Hiding the text file behind an image

1. Create a folder named Steganography on the desktop.
2. Copy an image file (for example, image.jpg and a text file (for example, secret.txt) into the folder.
3. Open the text file and enter the message to be hidden. Save and close it.
4. Open Command Prompt.
5. Navigate to the folder using the following command:

cd %USERPROFILE%\Desktop\Steganography

6. Use the following command to combine the image and text file:

copy /b image.jpg + secret.txt hidden.jpg

7. The combined file hidden.jpg will be created in the same folder.

Part B: Extracting the hidden text

8. Open Command Prompt in the same folder.
9. Use the following command to extract the appended text:

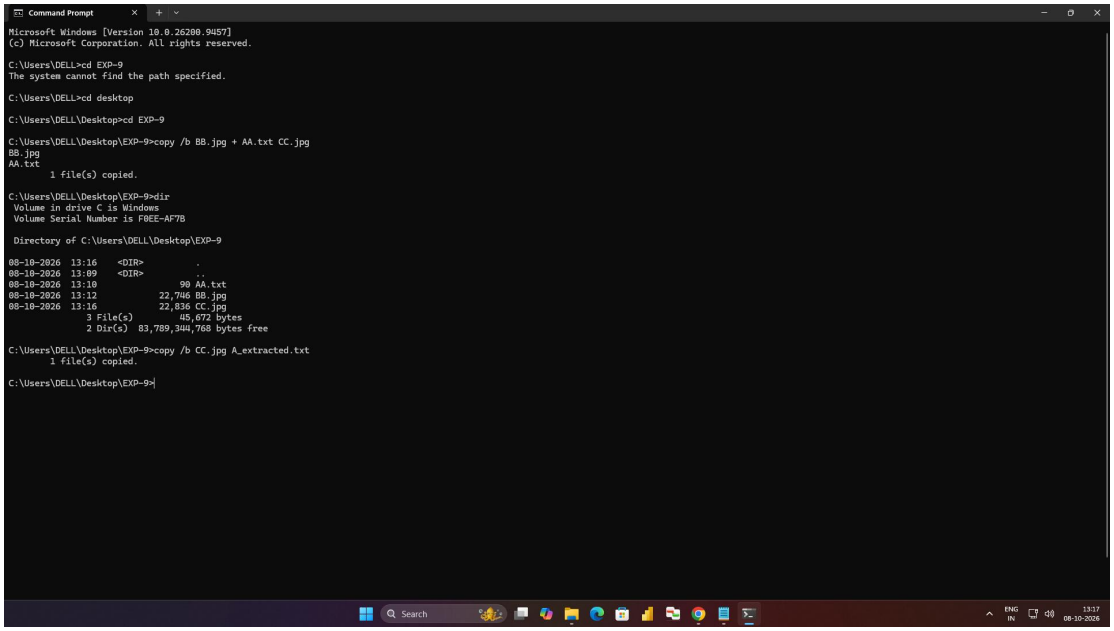
copy /b hidden.jpg A.txt

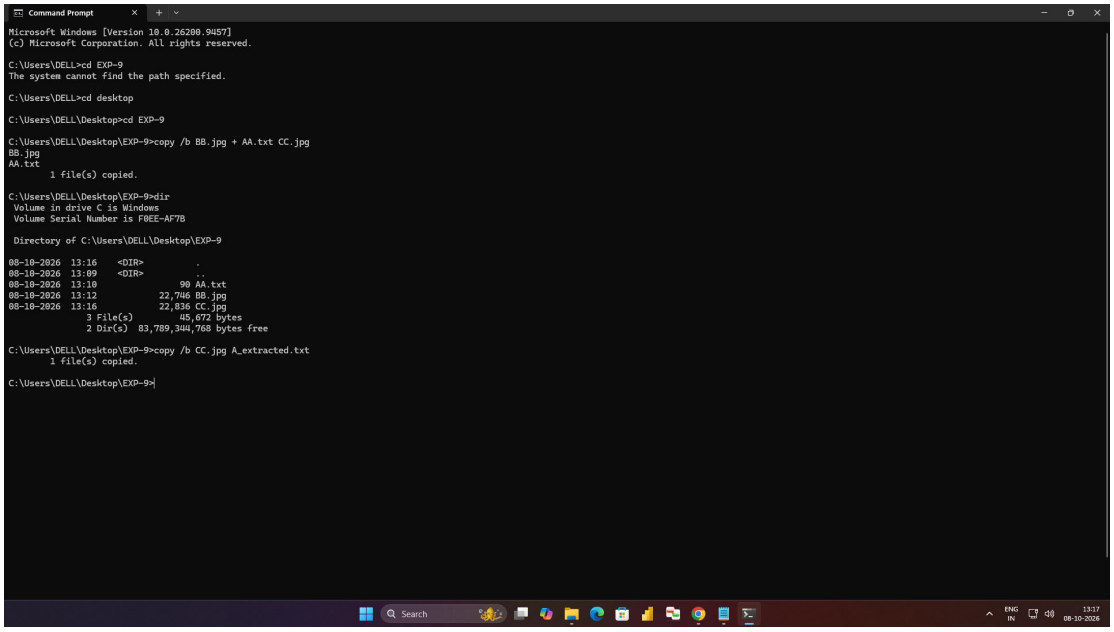
10. To recover the original text file more reliably, use a binary-aware editor or a small script to locate the appended text after the original image's end marker. For a JPEG file, the end marker is typically FF D9.

11. Save the recovered text into a new file named extracted.txt.

12. Open extracted.txt and verify that the hidden message has been recovered correctly.

OBSERVATIONS:





RESULTS:

Thus, a text file was appended to an image file using Windows Command Prompt, demonstrating a basic file-hiding technique. The appended text can be recovered using appropriate binary-aware extraction methods.

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HOW TO EXTRACT EXCHANGEABLE IMAGE FILE FORMAT (EXIF) DATA

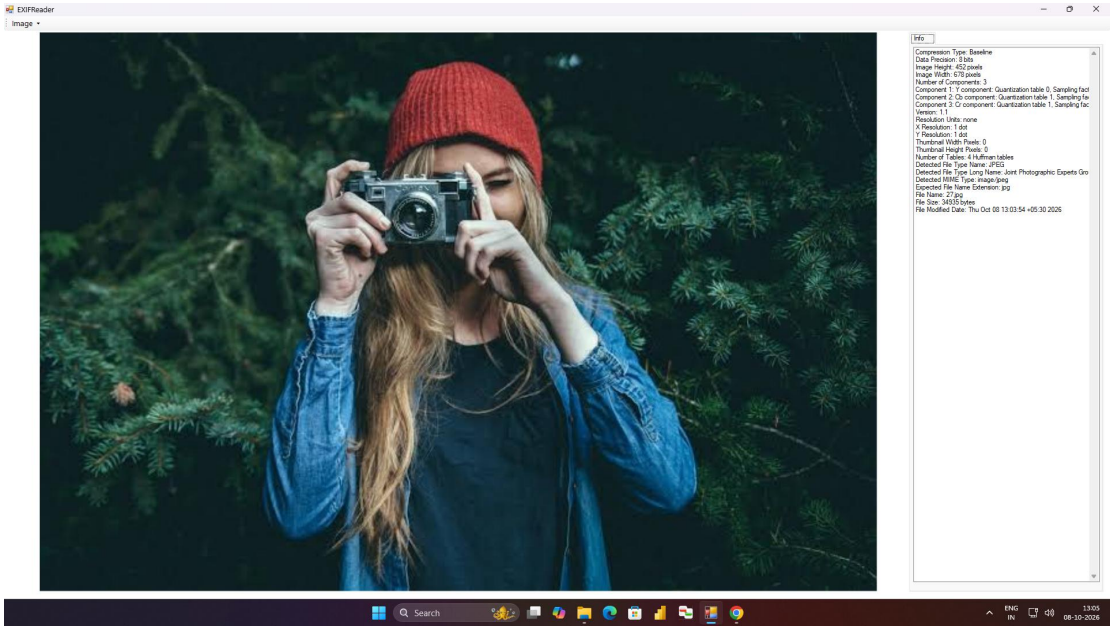
AIM:

To extract EXIF metadata from an image using the EXIF application and examine the details displayed by the application.

PROCEDURE:

1. Open the EXIF application on the computer.
2. Select the Open Image or Browse option in the application.
3. Navigate to the folder containing the image file.
4. Select the required image file and click Open.
5. The application displays the EXIF metadata associated with the selected image.
6. Observe the image details displayed in the application.
7. Examine the available information, such as camera make, camera model, date and time, image dimensions, resolution, and GPS location.
8. Note the metadata details displayed by the application.
9. Record the observations for forensic investigation.
10. Close the application after completing the analysis.

OBSERVATIONS:



RESULTS:

Thus, EXIF data was extracted from the selected image using the EXIF application, and the available image metadata was displayed and examined for forensic investigation.